COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S1
DATE PERFORMED:	July 9, 2026	**DATE SUBMITTED:**	July 14, 2026
NAME:	Nichollo Dave A. Galvez	**INSTRUCTOR:**	Sir JIMLORD QUEJADO

ACTIVITY NO.1 [ACTIVITY NAME] REVIEW OF C++ PROGRAMMING

1. PROCEDURE OUTPUT

What procedures did you perform?

1. **Create a function that will take two numbers and display the sum.**

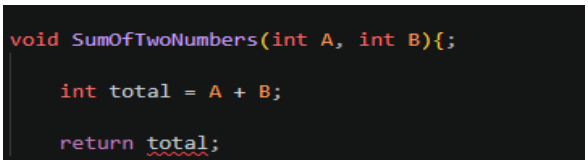
Input:

```
4   int SumOfTwoNumbers(int A, int B);
5
6
7
8   int main(){
9       int A,B;
10
11      namespace std      s your First number: "<<std::endl;
12      std::cin >> A;
13      std::cout<<"What is your Second number: "<<std::endl;
14      std::cin >> B;
15      std::cout<<"The sum is "<<SumOfTwoNumbers(A,B)<<std::endl;
16
17      return 0;
18
19  }
20
21
22
23  int SumOfTwoNumbers(int A, int B){;
24
25      int total = A + B;
26
27      return total;
28  }
```

Output:

```
What is your First number:
15
What is your Second number:
5
The sum is 20
PS C:\Users\TIPQC\CPE21s1\HOA_1\First>
```

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`void SumOfTwoNumbers(int A, int B);`
Class Declaration and Method Definition Section	None

Main Function	
Method Definition	

```cpp
#include <iostream>


void SumOfTwoNumbers(int A, int B);



int main(){
    int A,B;

    std::cout<<"What is your First number: "<<std::endl;
    std::cin >> A;
    std::cout<<"What is your Second number: "<<std::endl;
    std::cin >> B;
    std::cout<<"The sum is "<<SumOfTwoNumbers(A,B)<<std::endl;

    return 0;

}



void SumOfTwoNumbers(int A, int B){;

    int total = A + B;

    return total;
}
```

```cpp
void SumOfTwoNumbers(int A, int B){;

    int total = A + B;

    return total;
```

2. **Create a function that will return whether variable A is greater than variable B.**

Input:

```cpp
#include <iostream>

void Greater(int A, int B);

int main(){
    int A, B;

    std::cout<<"Enter your first number(A): ";
    std::cin>> A;
    std::cout<<"Enter your second number(B): ";
    std::cin>> B;

    Greater(A,B);

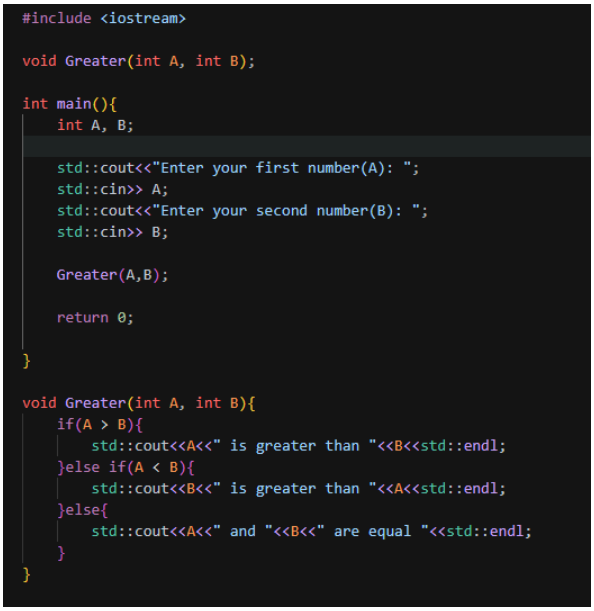
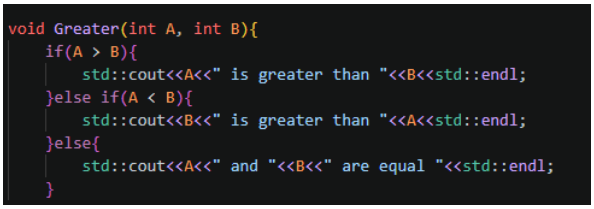
    return 0;

}

void Greater(int A, int B){
    if(A > B){
        std::cout<<A<<" is greater than "<<B<<std::endl;
    }else if(A < B){
        std::cout<<B<<" is greater than "<<A<<std::endl;
    }else{
        std::cout<<A<<" and "<<B<<" are equal "<<std::endl;
    }
}
```

Output:

```
Enter your first number(A): 5
Enter your second number(B): 5
5 and 5 are equal
PS C:\Users\TIPQC\CPE21s1\HOA_1\Second> C:\Users\TIPQC\CPE21s1\HOA
Enter your first number(A): 7
Enter your second number(B): 19
19 is greater than 7
PS C:\Users\TIPQC\CPE21s1\HOA_1\Second>
```

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`void Greater(int A, int B);`
Class Declaration and Method Definition Section	None
Main Function	
Method Definition	

```
#include <iostream>

void Greater(int A, int B);

int main(){
    int A, B;

    std::cout<<"Enter your first number(A): ";
    std::cin>> A;
    std::cout<<"Enter your second number(B): ";
    std::cin>> B;

    Greater(A,B);

    return 0;

}

void Greater(int A, int B){
    if(A > B){
        std::cout<<A<<" is greater than "<<B<<std::endl;
    }else if(A < B){
        std::cout<<B<<" is greater than "<<A<<std::endl;
    }else{
        std::cout<<A<<" and "<<B<<" are equal "<<std::endl;
    }
}
```

```
void Greater(int A, int B){
    if(A > B){
        std::cout<<A<<" is greater than "<<B<<std::endl;
    }else if(A < B){
        std::cout<<B<<" is greater than "<<A<<std::endl;
    }else{
        std::cout<<A<<" and "<<B<<" are equal "<<std::endl;
    }
```

COMPUTER ENGINEERING

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Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Input:

```cpp
#include <iostream>

bool logicalOperations(bool a, bool b)
{
    std::cout<< "\nLogical Operations\n";
    std::cout<< "a && b = " << (a && b) <<std::endl;
    std::cout<< "a || b = " << (a || b) <<std::endl;
    std::cout<< "!a = " << (!a) <<std::endl;
    std::cout<< "!b = " << (!b) <<std::endl;

    return true;
}

int main()
{
    bool a, b;
    bool Success;

    std::cout<< "Enter first Boolean value (0 or 1): ";
    std::cin>> a;

    std::cout<< "Enter second Boolean value (0 or 1): ";
    std::cin>> b;

    Success = logicalOperations(a, b);

    std::cout<< "\n Returned: " << Success <<std::endl;

    return 0;
}
```

Output:

```
Enter first Boolean value (0 or 1): 0
Enter second Boolean value (0 or 1): 1

Logical Operations
a && b = 0
a || b = 1
!a = 1
!b = 0

Function returned: 1
```

Sections	**Answer**
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`bool logicalOperations(bool a, bool b)`
Class Declaration and Method Definition Section	None

Main Function

```cpp
#include <iostream>

bool logicalOperations(bool a, bool b)
{
    std::cout<< "\nLogical Operations\n";
    std::cout<< "a && b = " << (a && b) <<std::endl;
    std::cout<< "a || b = " << (a || b) <<std::endl;
    std::cout<< "!a = " << (!a) <<std::endl;
    std::cout<< "!b = " << (!b) <<std::endl;

    return true;
}

int main()
{
    bool a, b;
    bool Success;

    std::cout<< "Enter first Boolean value (0 or 1): ";
    std::cin>> a;

    std::cout<< "Enter second Boolean value (0 or 1): ";
    std::cin>> b;

    Success = logicalOperations(a, b);

    std::cout<< "\n Returned: " << Success <<std::endl;

    return 0;
}
```

Method Definition

```cpp
bool logicalOperations(bool a, bool b)
{
    std::cout<< "\nLogical Operations\n";
    std::cout<< "a && b = " << (a && b) <<std::endl;
    std::cout<< "a || b = " << (a || b) <<std::endl;
    std::cout<< "!a = " << (!a) <<std::endl;
    std::cout<< "!b = " << (!b) <<std::endl;
```

Input:

```cpp
#include <iostream>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

bool Triangle::validateTriangle() {
    return (totalAngle == 180 && angleA > 0 && angleB > 0 && angleC > 0);
}
```

```cpp
int main() {

    Triangle set1(40, 30, 110);

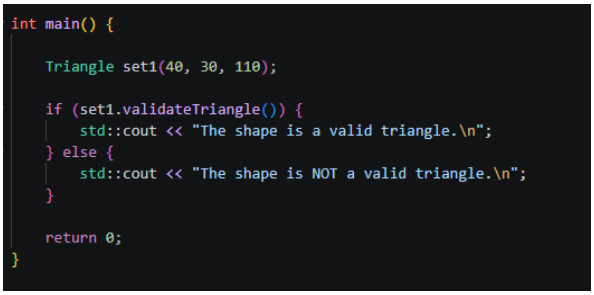
    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

Output:

```
The shape is a valid triangle.
```

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`void Triangle::setAngles(double A, double B, double C) { angleA = A; angleB = B; angleC = C; totalAngle = A + B + C; }`
Class Declaration and Method Definition Section	`class Triangle { private: double totalAngle, angleA, angleB, angleC; public: Triangle(double A, double B, double C); void setAngles(double A, double B, double C); bool validateTriangle();`

Main Function	 
Method Definition	

```cpp
#include <iostream>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

bool Triangle::validateTriangle() {
    return (totalAngle == 180 && angleA > 0 && angleB > 0 && angleC > 0);
}
```

```cpp
int main() {

    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

```cpp
bool Triangle::validateTriangle() {
    return (totalAngle == 180 && angleA > 0 && angleB > 0 && angleC > 0);
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a C++ program to swap the two numbers in different variables.

Input:

```cpp
#include <iostream>

void swapNumbers(int &a, int &b)
{
    int swap = a;
    a = b;
    b = swap;
}

int main()
{
    int num1, num2;

    std::cout << "Enter first number: ";
    std::cin >> num1;

    std::cout << "Enter second number: ";
    std::cin >> num2;

    std::cout << "\nBefore swapping:" << std::endl;
    std::cout << "First number: " << num1 << std::endl;
    std::cout << "Second number: " << num2 << std::endl;

    swapNumbers(num1, num2);

    std::cout << "\nAfter swapping:" << std::endl;
    std::cout << "First number: " << num1 << std::endl;
    std::cout << "Second number: " << num2 << std::endl;

    return 0;
}
```

Output:

```
Enter first number: 67
Enter second number: 76

Before swapping:
First number: 67
Second number: 76

After swapping:
First number: 76
Second number: 67
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

Input:

supp2 > main.cpp > main()

```cpp
#include <iostream>

double kelvinToFahrenheit(double kelv)
{
    return (kelv - 273.15) * 9 / 5 + 32;
}

int main()
{
    double k, f;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> k;

    f = kelvinToFahrenheit(k);

    std::cout << "Temperature in Fahrenheit: " << f << std::endl;

    return 0;
}
```

Output:

```
Enter temperature in Kelvin: 67
Temperature in Fahrenheit: -339.07
```

3. Create a C++ program that has a function that will calculate the distance between two points.

Input:

```cpp
#include <iostream>
#include <cmath>

double calculateDistance(double x1, double y1, double x2, double y2)
{
    return std::sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
}

int main()
{
    double x1, y1, x2, y2;

    std::cout << "Enter x1 and y1: ";
    std::cin >> x1 >> y1;

    std::cout << "Enter x2 and y2: ";
    std::cin >> x2 >> y2;

    double distance = calculateDistance(x1, y1, x2, y2);

    std::cout << "Distance = " << distance << std::endl;

    return 0;
}
```

Output:

```
Enter x1 and y1: 6
7
Enter x2 and y2: 8
9
Distance = 2.82843
```

4. Modify the code given in ILO B and add the following functions:

Input:

```cpp
#include <iostream>
#include <cmath>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;
    double sideA, sideB, sideC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    bool validateTriangle();

    void setSides(double a, double b, double c);
    double perimeter();
    double area();
    void triangleType();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;

    sideA = sideB = sideC = 0;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```cpp
bool Triangle::validateTriangle() {
    return (totalAngle == 180 && angleA > 0 && angleB > 0 && angleC > 0);
}

void Triangle::setSides(double a, double b, double c) {
    sideA = a;
    sideB = b;
    sideC = c;
}

double Triangle::perimeter() {
    return sideA + sideB + sideC;
}

double Triangle::area() {
    double s = perimeter() / 2;
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

void Triangle::triangleType() {
    if (angleA < 90 && angleB < 90 && angleC < 90)
        std::cout << "Triangle is Acute-angled.\n";
    else if (angleA > 90 || angleB > 90 || angleC > 90)
        std::cout << "Triangle is Obtuse-angled.\n";
    else
        std::cout << "Triangle is Others (Right triangle).\n";
}
```

Input:

```
int main() {

    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";

        // Enter the lengths of the sides
        set1.setSides(5, 6, 7);

        std::cout << "Perimeter: " << set1.perimeter() << std::endl;
        std::cout << "Area: " << set1.area() << std::endl;

        set1.triangleType();
    }
    else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

Output:

```
The shape is a valid triangle.
Perimeter: 18
Area: 14.6969
Triangle is Obtuse-angled.
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Assistance, Code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used it on how to put the formula of kelvin to fahrenheit. I request for help in supplementary number 4 on how to class works.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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